Poles of Multivariable Systems

Multivariable Control, Homework 4

Arshia Goshtasbi • MSc Electrical Engineering (Control), Iran University of Science and Technology • Autumn 1403 (2024)

English summary of my original Persian report, which is kept as a PDF in this folder.

- **The pole-polynomial rule.** I found the poles of

$$G(s) = \frac{1}{(s+1)(s+3)^2} \begin{bmatrix} s^2+6s+9 & -1 \\ 1 & s^2-9 \\ 2(s^2-9) & 1 \end{bmatrix}$$

  by reducing each entry, listing all non-zero minors, and taking the least common denominator. The pole polynomial is $(s+1)(s+3)^2$ (slide 17).

- **A second transfer matrix** (slide 27): its poles and their multiplicities.

- **Comparing pole-finding methods** (slide 47), from four representations:

  1. state space: the eigenvalues of a minimal realisation;
  2. the Rosenbrock system matrix: watch for cancellations;
  3. the transfer matrix: the LCM of the denominators of all minors;
  4. matrix-fraction descriptions: the determinant of the denominator matrix.